Illustration 1.13 By using the same concept applied in Illustration 1.12, graphs of $y = \sin x$ and $y = \sin 2x$, and also their combined graphs are given Figures 1.60, 1.61 and 1.62. The minimum and maximum values of $\sin x$ and $\sin 2x$ are the same. But they have different x-intercepts. The x-intercepts for $y = \sin x$ are $\pm n\pi$ and for $y = \sin 2x$ are $\pm\frac{1}{2}n\pi, n \in \mathbb{Z}$.

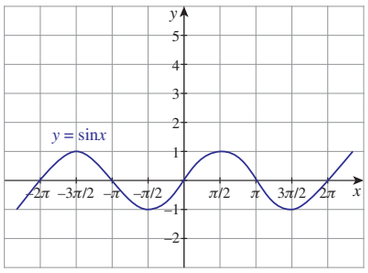


Figure 1.60

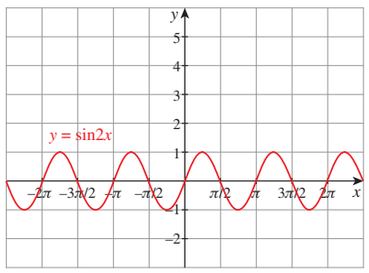


Figure 1.61

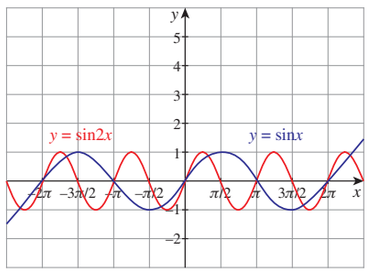


Figure 1.62

In the beginning of the section we talked about drawing the graph of $y = 2\sin(x - 1) + 3$. Now we are well equipped to draw the curve and even we can draw more complicated curve.

Illustration 1.14 Let us now draw the graph of $y = 2\sin(x - 1) + 3$.

It is clear that the curve can be obtained from that of $y = \sin x$ using translation and dilation.

So first we draw $y = \sin x$. From that it is easy to draw the curve $y = \sin(x - 1)$; then draw $y = 2\sin(x - 1)$ and finally $y = 2\sin(x - 1) + 3$. (See Figures 1.63 to 1.66.)

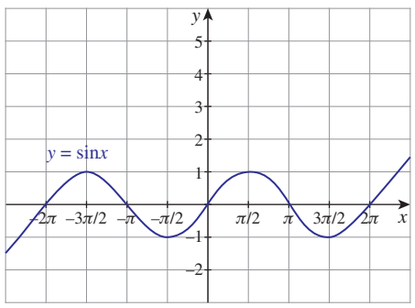


Figure 1.63

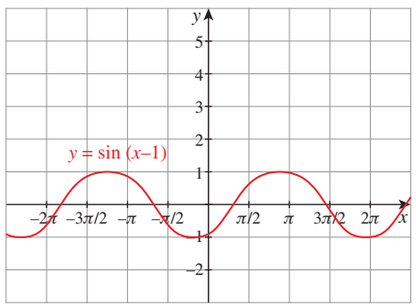


Figure 1.64

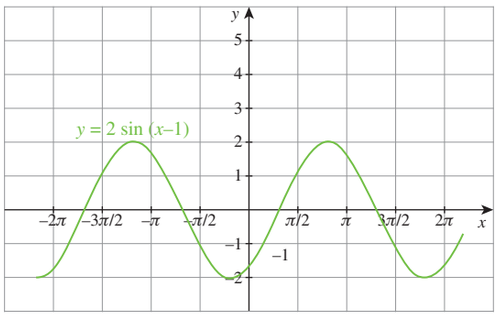


Figure 1.65

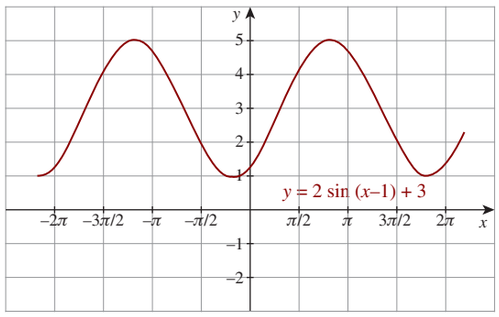


Figure 1.66